Modelling, Methodology and Consulting Skills (MMCS)

Optimisation Models for Renewable Energy Investment and Flexibility

MSc Operational Research - Project brief for academic year 2026/27

Introduction

Students will act as technical consultants engaged by a hypothetical Scottish renewable energy investment consortium. The client is considering investment in new renewable generation in Scotland, together with flexible assets that can absorb or shift electricity when renewable generation is not well aligned with demand and available routes for using that electricity.

The project combines strategic investment decisions with operational scheduling. Students must decide how a pre-defined infrastructure performs across changing demand, renewable generation and market conditions. There are two options for further development, either exploring investment in alternative infrastructure or building a temporal analysis which can allow the stakeholder project their investment over time. The main modelling work should be implemented in Python using the Xpress optimiser.

The project is intentionally open-ended. There is no single required objective function, model architecture or data processing route. A successful submission should demonstrate a robust modelling methodology, clear assumptions, appropriate use of data and public information, technical validation and a consulting style interpretation of the results.

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Client context

You are working as a team of technology consultants advising a hypothetical organisation considering significant investment in renewable energy infrastructure. The client has already received a proposed solution from another consultancy but was not satisfied that it provided a sufficiently convincing investment strategy. Your task is to scrutinise the problem independently, challenge the assumptions behind a simple solution and develop an improved strategy supported by quantitative analysis.

The central question is:

What combination of renewable generation and flexible energy technologies should the client invest in and how should those assets be used to create a financially and operationally sustainable strategy?

This is a combination of investment and optimisation problem, hence do not consider it as a detailed electrical engineering exercise. Your recommendations should be understandable to a senior investor as well as technically defensible. The project concept combines strategic investment choices with decisions about how infrastructure should be operated and expects the modelling to be implemented in a clear technical way.

For the purposes of this project, treat the infrastructure using the supplied datasets as the client's existing position. This represents the starting point inherited from the previous consultancy. Your role is to determine how effectively that infrastructure can be used and then to demonstrate whether a more sophisticated strategy materially improves the client's position. The supplied generation, demand and pricing data represent a constructed energy system and should not be interpreted as a detailed picture of Scottish transmission constraints or actual curtailment events.

Data provided

You will receive five core datasets as a starting point.

- *daily_generation_by_fuel.csv* contains daily **electricity generation** by fuel type – including wind, solar, hydro, nuclear, gas and other generation technologies.
- *daily_generation_total.csv* - total **daily generation**. This is the aggregate of the fuel level generation data.
- *daily_demand.csv* - daily **electricity demand** over several years (since 2020), allowing you to examine both typical conditions and periods when supply and demand are more challenging.
- *wind_farm_generation.csv* - the daily **generation profile** associated with the wind farm infrastructure used in the baseline case.
- *daily_market_index_price.csv* - historical daily **wholesale electricity value**, expressed in £/MWh.

The datasets cover broadly the same period from 2020 onwards. You should begin by understanding and checking the data, prior to constructing the design and development of an optimisation model. Consider whether dates align correctly, whether observations are missing or unusual, how generation varies by technology and when renewable production is high or low relative to demand. Document any data adjustments you would like to propose.

You may supplement the input data with clearly referenced public information where this improves your model, particularly for technology investment costs or scenarios. The supplied datasets are intended to allow you to focus on the initial modelling, analysis and work towards consulting recommendation.

Your modelling challenge

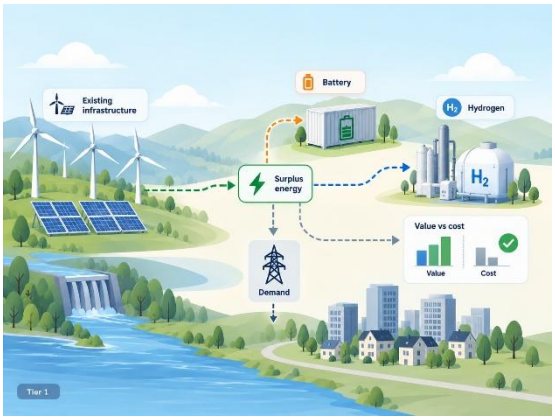
There are **two stages** to the modelling challenge. All teams should first develop the Tier 1 foundational (vanilla) model. Once this has been implemented, validated and interpreted, teams should select one of two extension routes: Tier 2A or Tier 2B. Tier 2A focuses on changing the infrastructure and investment

portfolio, while Tier 2B focuses on improving the way the existing portfolio is operated and assessed over time. Teams are not expected to complete both Tier 2 options. Depth, justification and interpretation of the chosen extension are more important than attempting to include every possible feature.

Tier 1 (mandatory) - Making use of surplus renewable energy

The first step is to recognise that renewable electricity is not always produced when it is most useful. At some points there may be more renewable generation available than can immediately be used.

Using the infrastructure and historical generation profiles represented in the supplied data, build a foundational optimisation model that identifies when electricity is available beyond immediate demand and investigates whether **battery storage** and/or **hydrogen production** can make better use of it. Your principal investment decisions should relate to how much battery and/or electrolyser capacity is worth adding and how much otherwise unused renewable energy these technologies can capture.



For battery storage, use a standard 2-hour battery. This means that, for example, a 50 MW battery can store 100 MWh of electricity. Assume 90% roundtrip efficiency and that normally only 20%-80% of the battery's total capacity is used. As an indicative investment assumption, use €0.288 million per MWh of storage capacity. The amount of battery capacity to build should be a decision variable.

For hydrogen, you may consider a PEM electrolyser. A commercial 20 MW ITM Power POSEIDON system can be used as a reference but again – the amount built should be a decision. Assume that the electrolyser can operate between 20% and 100% of installed capacity and requires approximately 55.9 kWh of electricity to produce 1 kg of hydrogen. In plain terms, surplus renewable electricity can therefore be converted into a product with potential economic value.

You should investigate and justify an appropriate electrolyser investment cost. For an initial economic scenario, you may use approximately £9.5/kg of hydrogen as a benchmark but you should recognise that this is based on a UK policy supported hydrogen value (rather than a guaranteed commercial selling price). Testing lower and higher hydrogen values is therefore encouraged.

The key client question at this level is:

Given the existing generation infrastructure, how much battery or hydrogen capacity should the client invest in and is the additional value created sufficient to justify that investment?

Tier 1 models should use a simplified treatment of time – detailed chronological operation of storage is not required at this stage.

Tier 2 (option A) - Generation investment

The key question here is whether the client should change the **generation infrastructure** assumed in the baseline case. Extend your foundational model by considering an alternative generation infrastructure. Using the historical data as evidence of how different generation technologies behave, determine what generation capacity the investor should support or develop, should they decide to invest in new infrastructure. A useful starting point may be to explore in-development projects across Scotland and the wider UK area (a recommended source as a starting point: Renewable Energy Planning Database: quarterly extract - GOV.UK).

Your model should answer questions such as:

- Which technologies should the client invest in?
- How much capacity should be developed?
- What level of demand can the proposed portfolio serve?
- How much renewable generation could be produced but not usefully consumed?

Investment cost should be part of the decision. You will therefore need to compare the additional energy or benefit created by an investment with the amount of capital required.



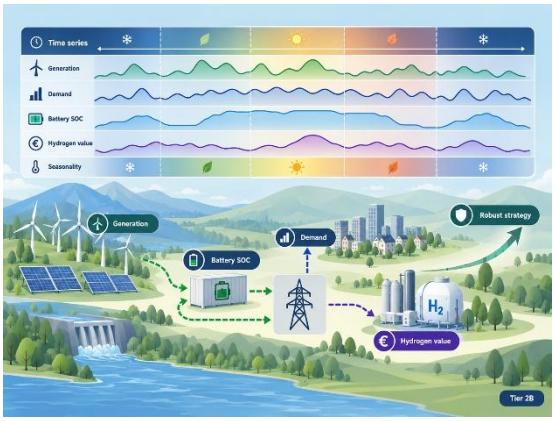
This provides the client's infrastructure investment strategy and a benchmark against which sophisticated solutions can be compared. Your optimisation should distinguish between the infrastructure already represented in the supplied data and new investment decisions introduced by your team. A strong Tier 2A analysis will show the marginal benefit of additional infrastructure, identify where diminishing returns arise and compare additional generation investment with further investment in battery or hydrogen capacity.

Tier 2 (option B) - A sustainable strategy through time

In the consulting industry, a convincing solution presents how strategy and decision making are influenced over time, i.e. make use of **time series**.

In the Tier 2B extension you should explicitly preserve the chronology of the supplied data. Unlike the Tier 1 approximation, decisions on one day should influence what is possible on later days. For example, a battery should have a state of charge that carries from one period to the next and electricity or hydrogen value may vary with market conditions. Teams should investigate seasonal behaviour, prolonged periods of high or low renewable generation and unusually high demand or market prices.

A battery cannot discharge electricity today unless energy has previously been stored. Similarly, the value of renewable generation depends on when it is available relative to demand. An investment that appears attractive when annual averages are considered may perform poorly during periods of low renewable output, high demand or prolonged surplus generation.



Your advanced model should therefore determine how the investment portfolio performs **day by day over time**. It should identify how the assets should be operated.

As defined earlier, the client ultimately wants to understand what a successful and sustainable strategy looks like. Sustainability here should be considered broadly:

- sensible use of investment capital, i.e. return on investment (ROI) is a factor for every investor
- good use of renewable energy – in the context of the UK and global sustainability goals
- resilience to changing conditions, i.e. a recommendation that remains flexible, not dependent on a particular set of assumptions.

You might therefore test alternative levels of demand, renewable generation, battery costs, hydrogen values or investment budgets. You should identify which assumptions materially change the recommended strategy. **The objective** could be, for example, to:

- minimise cost
- maximise renewable utilisation
- minimise unused generation
- maximise economic value
- or balance several of these considerations.

The important requirement is that you explain why your chosen objective (or a combination of multiple) represents the client's interests.

What the client expects

Your final recommendation should go beyond reporting the output of an optimisation models. The client wants to know what to invest in, how much to invest and why the recommendation is better than a simpler strategy. Furthermore, a very important question of interest is what could cause the recommendation to change.

You should therefore demonstrate that you have challenged your own model.

- ✓ Check that the results make sense and display appropriate visualisations.
- ✓ Compare the optimised solution with simple alternatives.
- ✓ Explain which assumptions have the greatest influence through sensitivity analysis.
- ✓ Distinguish clearly between conclusions supported by the data and assumptions introduced for modelling purposes.

You will have access to supervision throughout the project. However, the modelling choices, validation and recommendation remain the responsibility of your team.

A successful project will tell a clear consulting story: understand the problem, establish a credible baseline, improve the use of renewable energy through investment and flexibility, stress test the strategy over time (if Tier 2B is chosen) and give the client a concise recommendation about where its money should go.

Appendix

Battery and electrolyser parameters and recommended sources:

Subject area	Recommendation	Notes	Source
Battery size	2-hour battery	If 50 MW of battery capacity is built, assume it can store 100 MWh. Installed MW should be a decision variable.	Tesla Megapack
Battery efficiency	90% round-trip efficiency	This allows for some electricity being lost when energy is stored and later released.	
Battery usable range	20%-80% of total capacity	As a modelling assumption, prevent the model from repeatedly using the battery from completely empty to completely full.	
Battery investment cost	€0.288m per MWh	Multiply installed battery energy capacity by this figure to estimate investment cost.	Danish Energy Agency - Energy Storage (the data sheet)
Electrolyser type	PEM (can be another type)	Treat this as flexible electricity demand that produces hydrogen when electricity is available.	ITM Power POSEIDON
Electrolyser scale	20 MW reference, but capacity should be a decision variable	20 MW provides a realistic commercial benchmark, whilechoosing the amount worth building.	
Electrolyser operating range	20%-100% of installed capacity	Hence, a 20 MW electrolyser would normally operate between 4 MW and 20 MW when running.	
Hydrogen production	The electrolyser needs about 55.9 kWh of electricity to produce 1 kg of hydrogen.	Convert electricity used by the electrolyser into kilograms of hydrogen produced.	
Hydrogen value	£9.5/kg as a core scenario £241/MWh x 39.4 kWh/kg / 1000 = £9.5/kg (see the sources in the last column)	Use this to estimate the value of hydrogen produced, but also test lower and higher values. It is a supported benchmark, not a guaranteed market price.	Net Zero Hydrogen Fund: HAR1 successful projects (published December 2023) - GOV.UK For weighted average strike price of £241/MWh Hydrogen Transport and Storage Cost Report For energy content of hydrogen 39.4 kWh/kg